

CLO1 Distinguish the basic problem-solving techniques in developing algorithms and programs for given problems. [C4] - 1%						
Criteria / Marking Scale	0	0.25	0.5	0.75	1	Mark
Ability to analyze given problem and identify correct requirement	Unable to identify any correct input, process and output from a given problem	Able to identify partially correct input / process / output	Able to identify basic requirement for input, process and output from a given problem	Able to identify correctly all input, process and output from a given problem	Able to identify correctly all input, process and output from a given problem, and provide alternative	/1
CLO2 Construct complete programs based on a given specification. [P4] - 5%						
Criteria / Marking Scale	0	0.25	0.5	0.75	1	
Ability to apply good programming practices in program – indentation, comments, variables	Unable to organize the code	The code is poorly organized and difficult to read	The code is readable only by a person who already knew the code purposes	The code is organized and easy to read	The code is well organized, easy to read and provide proper comment to ease code understanding	/1
Ability to run/debug	Unable to run the program (syntax error)	Able to run the program but no results produced	Able to run the program successfully but produced wrong results	Able to run the program successfully with partially correct results	Able to run the program successfully with correct results	/1
Input Validation	No validation at all	some validation with no informative msg	some validation with some informative msg	Perform most input validation with informative msg + limited snapshot	Perform most input validation with informative msg + clear snapshot	/1
Ability to produce a correct programming code based on programming design	Unable to provide the programming source code	Able to produce a programming source code but did not aligned with flowchart	Able to produce a programming source code but partially aligned with flowchart	Able to produce a programming source code but almost aligned with flowchart	Able to produce a programming source code that aligned with flowchart	/1

Screenshot for program features (Font size for snapshot approx. to Century Schoolbook, 11pt)	No screenshot submitted	Incomplete screenshots with unreadable fonts	Screenshots proper font size, missing screenshots for some program features	Complete screenshots with proper font size cover all program features but missing descriptions for some snapshots.	Complete screenshots with proper font size cover all program features. Proper descriptions are provided for each snapshot	/1
CLO3 Construct the process of top-down, stepwise refinement to benefit all related software construction task. [P4] – 4%						
Criteria / Marking Scale	0	0.25	0.5	0.75	1	
Ability to demonstrate good design solution – flowchart (standard and representation)	Unable to construct the flowchart design	Able to construct flowchart design but did many mistakes on symbols and flows and keywords	Able to construct the flowchart design with little mistake and incomplete solution	Able to construct the flowchart design correctly but incomplete solution	Able to construct the flowchart design correctly with complete solution	/2
Criteria / Marking Scale	0	0.5	1	1.5	2	
Ability to demonstrate good design solution and apply appropriate control structure (logic and control structure)	Only produce sequential solution	only able to apply selective conditions in solution	Able to construct acceptable logics in solutions but Did Not fulfill the requirement of 1 sequential AND 2 repetition structures in solution	Able to construct Acceptable logics in solutions and applied at least 1 sequential AND 2 repetition structures in solution	Able to construct great logics in solutions and applied at least 1 sequential AND 2 repetition structures in solution	/2
TOTAL						/4